

6-mavzu. Chiziqli tenglamalar sistemasi. Chiziqli tenglamalar sistemasini Gauss va teskari matritsa usuli yordamida echish.(2 soat)

Reja:

1. Chiziqli tenglamalar sistemasi.
2. Chiziqli tenglamalar sistemasini , Gauss usuli yordamida yechish.
3. Chiziqli tenglamalar sistemasini teskari matritsa yordamida yechish

Adabiyotlar. 2, 5, 6, 11,16.18

Tayanch iboralar: sistema koeffitsienti, ozod had, sistemaning yechimi, birgalidagi sistema, aniq sistema, aniqlas sistema, birgalikda bo'lmagan sistema.

Shuni aytish joizki, sistemadagi noma'lum (tenglama)lar soni orta borgan sari uni Kramer usuli bilan yechish qiyinlasha boradi.

Chiziqli tenglamalar sistemasini matritsali yozilishi va yechilishi. Osonlik uchun uch noma'lumli uchta chiziqli tenglamalar sistemasi

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = b_1 \\ a_{21}x + a_{22}y + a_{23}z = b_2 \\ a_{31}x + a_{32}y + a_{33}z = b_3 \end{cases}$$

(3.2.1)

ni qaraymiz.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

belgilashni kiritamiz. U holda matritsalarini ko'paytirish qoidasiga binoan

$$A \cdot X = \begin{pmatrix} a_{11}x + a_{12}y + a_{13}z \\ a_{21}x + a_{22}y + a_{23}z \\ a_{31}x + a_{32}y + a_{33}z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

bo'ladi. Shuning uchun (5.1) sistemani matritsalarini tengligidan foydalanib

$$\begin{pmatrix} a_{11}x + a_{12}y + a_{13}z \\ a_{21}x + a_{22}y + a_{23}z \\ a_{31}x + a_{32}y + a_{33}z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

ko'rinishida yoki qisqacha $A \cdot X = B$ (3.2.2) matritsali tenglama ko'rinishida yozish mumkin. (3.2.1) sistema (3.2.2) yordamida berilgan bo'lib A xosmas matritsa bo'lsa uni quyidagicha yechiladi. (3.2.2) tenglamaning har ikkala tomonini chapdan A matritsaga teskari A^{-1} ga ko'paytirsak, $A^{-1} \cdot (A \cdot X) = A^{-1} \cdot B$; $(A^{-1} \cdot A)X = A^{-1} \cdot B$, $E \cdot X = A^{-1}B$; $X = A^{-1} \cdot B$.

$X = A^{-1} \cdot B$ (3.2.3) matritsali tenglama (3.2.2) ning yechimidir.

1-misol.

$$\begin{cases} 2x - y + z = 3 \\ x + 2y - 2z = -1 \end{cases}$$

sistema matritsa usulida yechilsin.

Yechish. Bu misolda

$$A = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

bo'lgani uchun berilgan sistemani $A \cdot X = B$ matritsali ko'rinishda va yechimini $X = A^{-1}B$ ko'rinishda yozish mumkin. A matritsaning determinanti 1-satr elementlarini 2 ga ko'paytirib 2-satrning mos elementlariga qo'shamiz va hosil bo'lgan determinantni ikkinchi satr elementlari bo'yicha yoyib hisoblaymiz.

$$|A| = \begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -2 \end{vmatrix}$$

bo'lgani uchun A xosmas matritsa va unga teskari A^{-1} matritsa mavjud. A matritsaning barcha elementlarini algebraik to'ldiruvchilarini topamiz:

$$\begin{aligned} A_{11} &= (-1)^{1+1} \begin{vmatrix} 2-2 \end{vmatrix} = -2, & A_{12} &= (-1)^{1+2} \begin{vmatrix} 1-2 \end{vmatrix} = -3, & A_{13} &= (-1)^{1+3} \begin{vmatrix} 12 \end{vmatrix} = -4, \\ A_{21} &= (-1)^{2+1} \begin{vmatrix} -11 \end{vmatrix} = -1, & A_{22} &= (-1)^{2+2} \begin{vmatrix} 21 \end{vmatrix} = -9, & A_{23} &= (-1)^{2+3} \begin{vmatrix} 2-1 \end{vmatrix} = -7, \\ A_{31} &= (-1)^{3+1} \begin{vmatrix} -11 \end{vmatrix} = 0, & A_{32} &= (-1)^{3+2} \begin{vmatrix} 21 \end{vmatrix} = 5, & A_{33} &= (-1)^{3+3} \begin{vmatrix} 2-1 \end{vmatrix} = 5. \end{aligned}$$

Algebraik to'ldiruvchilarning qiymatlarini (3.2.2) formulaga qo'yib A^{-1} teskari matritsani topamiz:

$$A^{-1} = \begin{pmatrix} -2 & -10 \\ -3 & -95 \end{pmatrix} \cdot \frac{1}{5}$$

X , A^{-1} va B matritsalarini $X = A^{-1} \cdot B$ tenglikga qo'yamiz. U holda

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -2 & -10 \\ -3 & -95 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ -6+1+0 \\ -9+9-10 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -5 \\ -10 \\ -5 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix}$$

hosil bo'ladi. Bundan $x=1, y=2, z=3$ echimni hosil qilamiz.